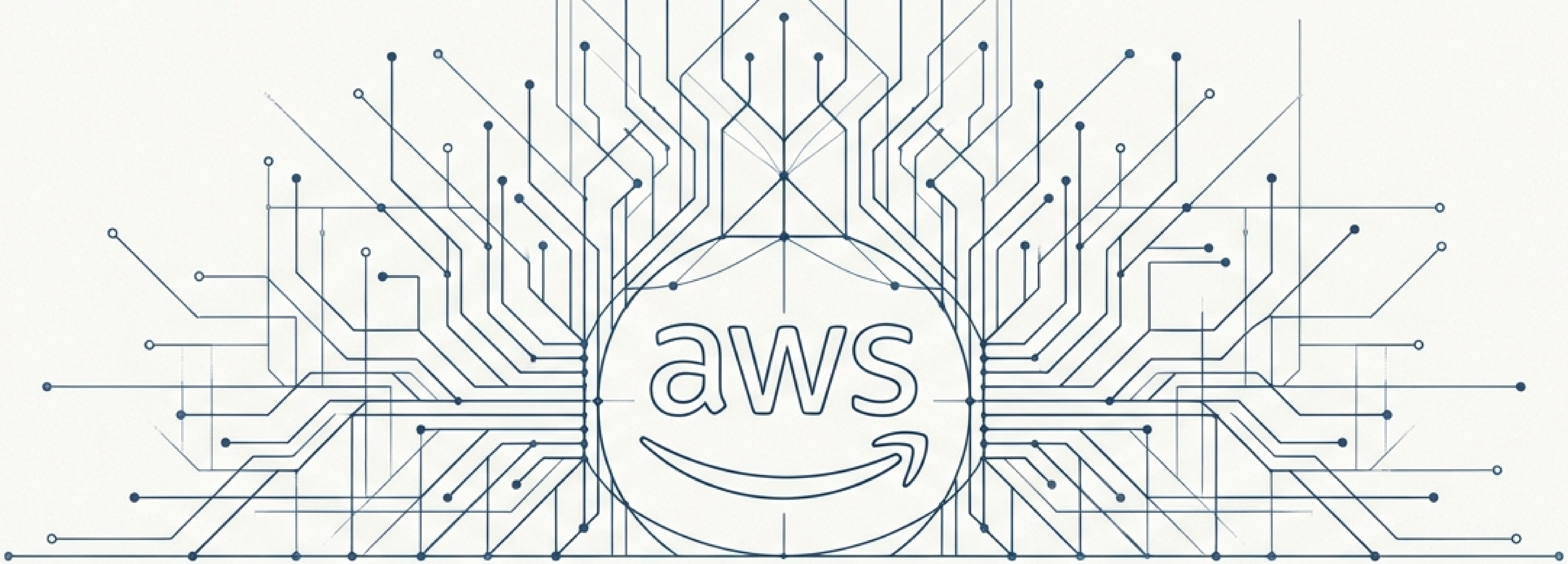


A Strategic Partnership to Advance Serverless and GenAI on AWS

Unlocking New Capabilities with the Innovations of Dinis Cruz

An Opportunity to Bridge Strategic Gaps in the AWS Ecosystem

For over six years, cybersecurity expert and innovator Dinis Cruz has developed a suite of cutting-edge solutions exclusively on AWS, creating technologies that fill critical gaps and showcase the full potential of serverless and GenAI. This proposal outlines a strategic partnership to integrate and amplify these innovations for mutual benefit.

The Innovator & The Vision

Dinis Cruz, a seasoned cybersecurity technologist, has a proven track record of building serverless-first, AI-driven applications on AWS.

The Proven Technology

His portfolio includes **OSBot-AWS** (a mature cloud automation toolkit), **MGraph-AI** (a novel serverless graph database), and **The Cyber Boardroom** (a GenAI platform with plans to integrate Amazon Bedrock).

The Strategic Partnership

We propose AWS co-develops these solutions, features them as flagship case studies, and sponsors their open-source evolution to secure a competitive advantage in the serverless and AI markets.

An Innovator with Deep, Long-Term AWS Expertise

- **Cybersecurity Expert & Software Innovator:** Founder of The Cyber Boardroom, Chief Scientist at Glasswall, and former OWASP leader.
- **6+ Years of AWS-Centric Development:** Extensive, hands-on proficiency building exclusively on AWS since 2018.
- **Early Adopter and Advocate of Serverless:** Recognised the agility and cost-efficiency of AWS Lambda, S3, and API Gateway to build highly scalable, flexible workflows.
- **Secure and Scalable by Design:** Background in application security informs the architecture of all solutions, emphasising security and enterprise-grade design on AWS.

A Six-Year Journey of Compounding Innovation on AWS

2019
OWASP Security Bot (OSBot)
Pioneered serverless workflows to create knowledge graphs from security data like Jira.

2018–2021
OSBot-AWS Toolkit
Expanded OSBot into an open-source Python library to simplify AWS automation over the boto3 SDK.

2022
GenAI & Knowledge Graph R&D
Began integrating LLMs and developing the core concepts for a memory-first, AI-native graph database on AWS.

Early 2024
The Cyber Boardroom Launch
Launched a fully serverless, GenAI-driven platform on AWS, built to serve executives.

Late 2024
Serverless News Feed Architecture
Engineered a high-performance, dual-API pipeline on Lambda and S3 for real-time data processing.

Early 2025
MGraph-AI Release
Open-sourced a purpose-built serverless graph database to fill a critical gap in the ecosystem.

March 2025
MyFeeds.ai MVP
Demonstrated a full-circle application of MGraph-AI and LLMs for personalised, graph-backed news feeds.

OSBot-AWS: The Foundation of Serverless Automation and Graph-Based Insights

An open-source Python toolkit maintained for over 6 years that provides a high-level wrapper around boto3, simplifying the automation of AWS resources (S3, EC2, Lambda, IAM).

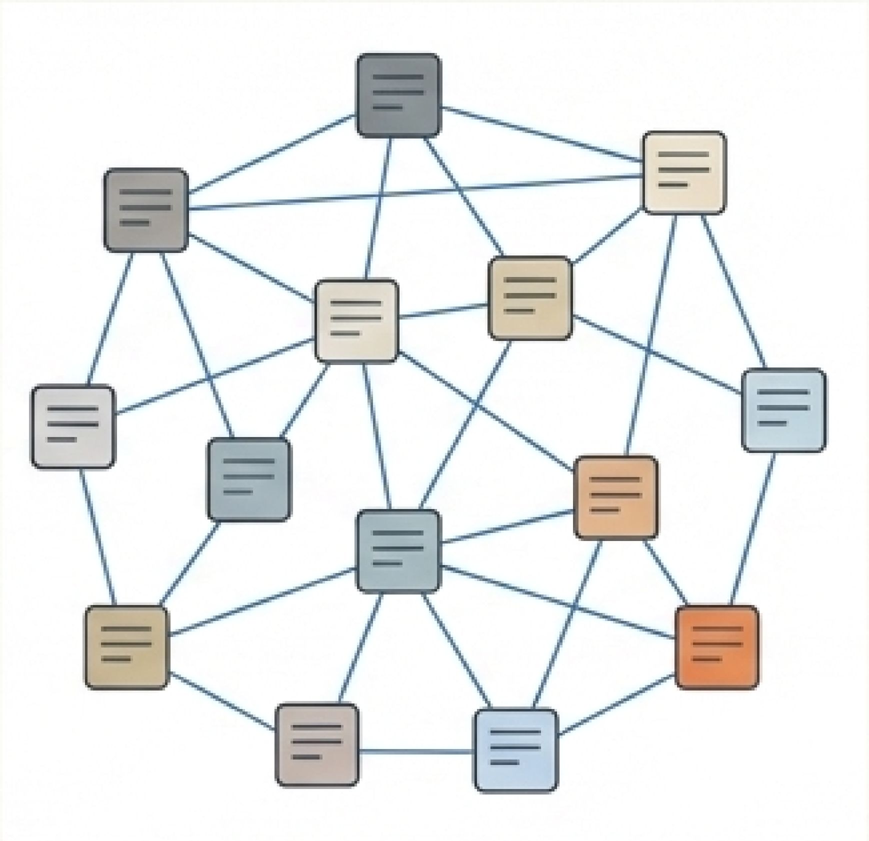
Key Innovation: 'Jira as a Graph Database'

Before

- Showcased in 2019, OSBot demonstrated how to treat Jira tickets and their relationships as nodes and edges in a knowledge graph.
- The system used a serverless architecture—AWS Lambda, Elastic, Slack—to pull data and generate visual maps of an organisation's security posture.
- This approach was ahead of its time, proving the value of on-demand graph analysis for IT data without dedicated servers.

Serverless Workflow

After



> *"generate graphs of data from tools like Jira to help understand security issues and their relationships."*

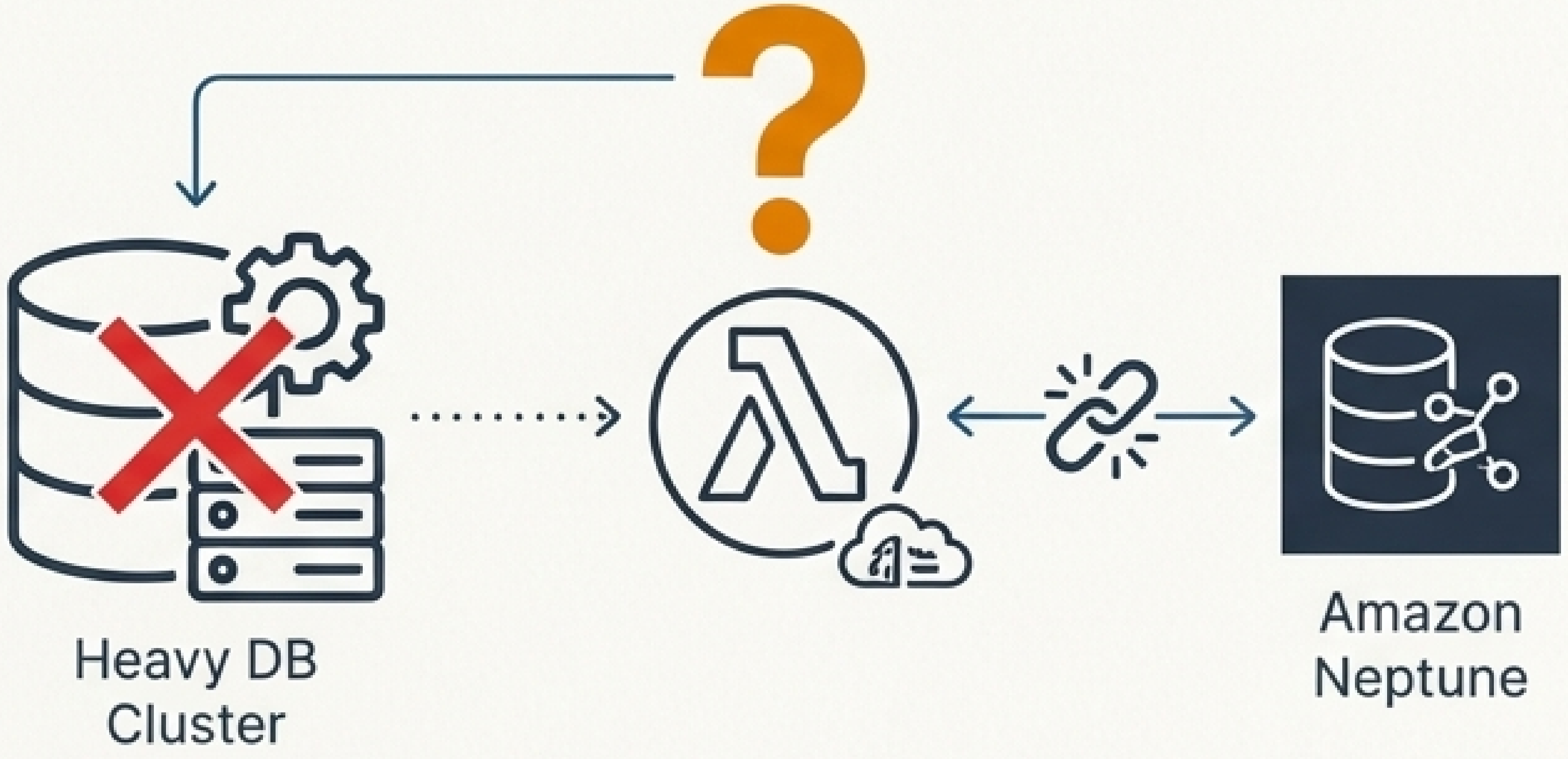
Identifying a Strategic Gap: The Need for a Truly Serverless Graph Database

The Core Problem

Developers building on AWS Lambda face a challenge when needing graph capabilities for transient, on-demand workloads.

Limitations of Existing Solutions

- **Traditional Graph Databases:** Too heavy, complex to deploy, and not cost-effective for ephemeral, on-demand use cases.
- **Amazon Neptune / Neptune Serverless:** Powerful for managed clusters but not designed for lightweight, in-process graph logic embedded directly within a Lambda function. The architecture is still based on database scaling, not zero-footprint execution.



> *"Traditional graph databases did not support serverless and were either too heavy, too complex to deploy, or just didn't fit my workflow."* - Innovator

MGraph-AI: A Memory-First Graph Database Designed for the Serverless Era

An open-source (Apache 2.0) Python graph database purpose-built for AI and serverless environments.

Memory-First Performance

Entire graph resides in memory for lightning-fast reads, writes, and traversal, ideal for latency-sensitive Lambda invocations.

JSON Persistence

Persists data to simple JSON files on disk or cloud storage (e.g., Amazon S3) for durability and easy integration.

Zero Cost When Idle

Perfectly aligned with the serverless cost model. No resources are consumed when not in use.

Optimised for GenAI

Designed specifically for workloads like retrieval-augmented generation (**Graph-RAG**), providing structured context to LLMs.

> Fills the gap for a lightweight, on-demand graph engine that can run inside Lambda or Fargate, a capability currently missing from the AWS portfolio.

In Practice: The Cyber Boardroom, a Live Serverless GenAI Platform on AWS

Platform Overview

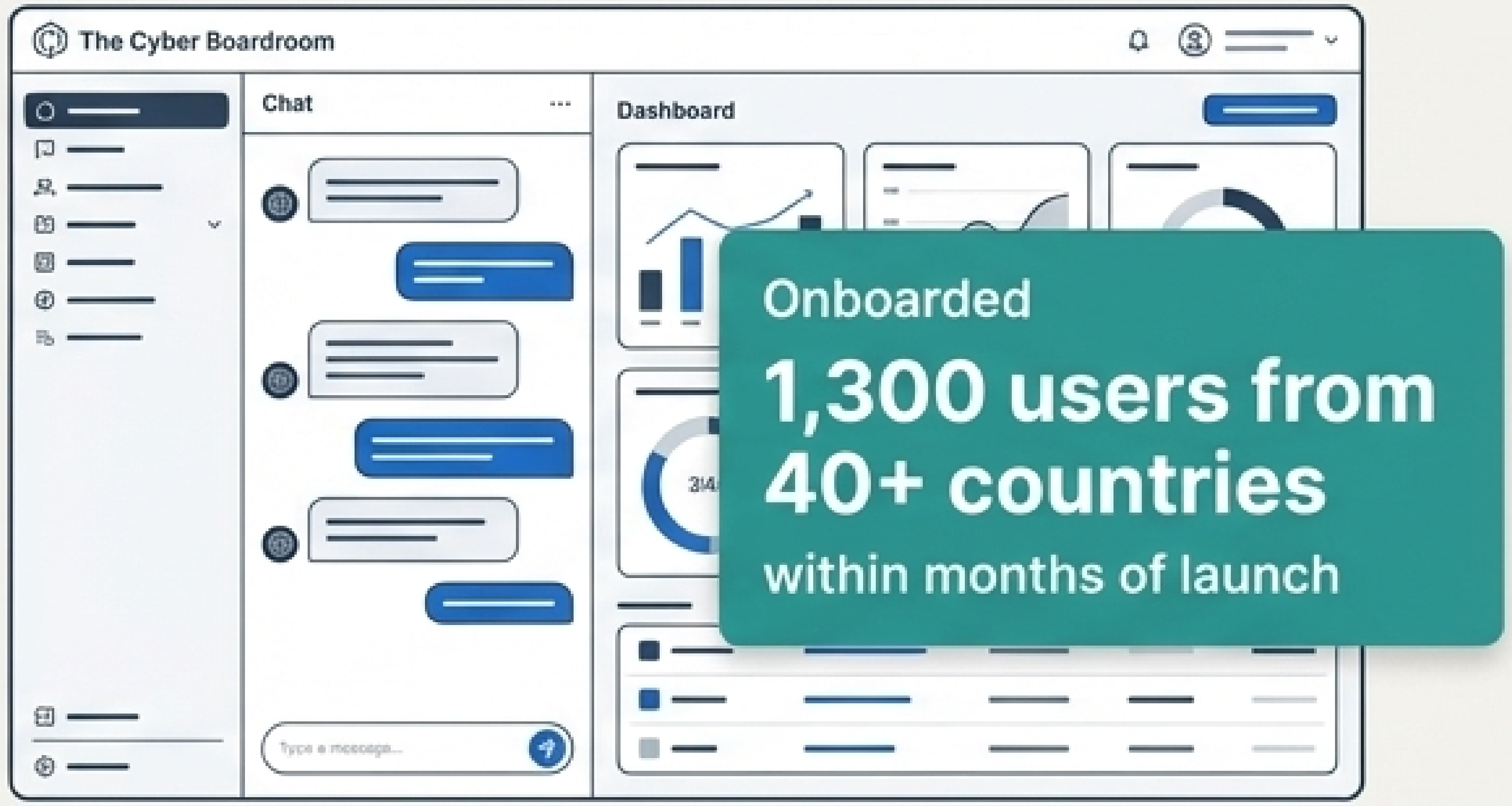
A cloud-native, serverless application providing personalised cybersecurity advice to executives and board members, powered by GenAI bots.

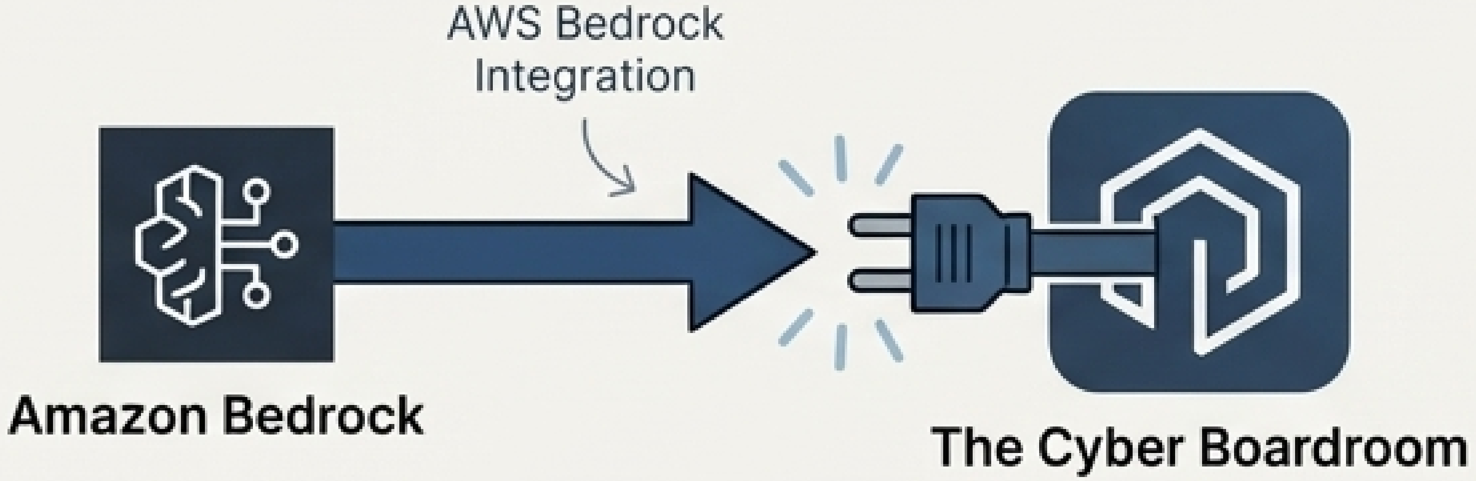
Built on AWS Serverless

- The AI assistant 'Athena' and all back-end logic run on AWS Lambda and API Gateway.
- Content and state are managed with Amazon S3 and other AWS storage services.
- The architecture allows for massive scalability with minimal infrastructure management.

Strategic Alignment with Amazon Bedrock

- Initially built with OpenAI's GPT-4, the platform is strategically designed to integrate with **Amazon Bedrock**.
- This positions The Cyber Boardroom as a prime candidate for a Bedrock success story, showcasing its use in a secure, enterprise-grade application.





MyFeeds.ai: Enhancing GenAI with Graph-Powered Context and Provenance

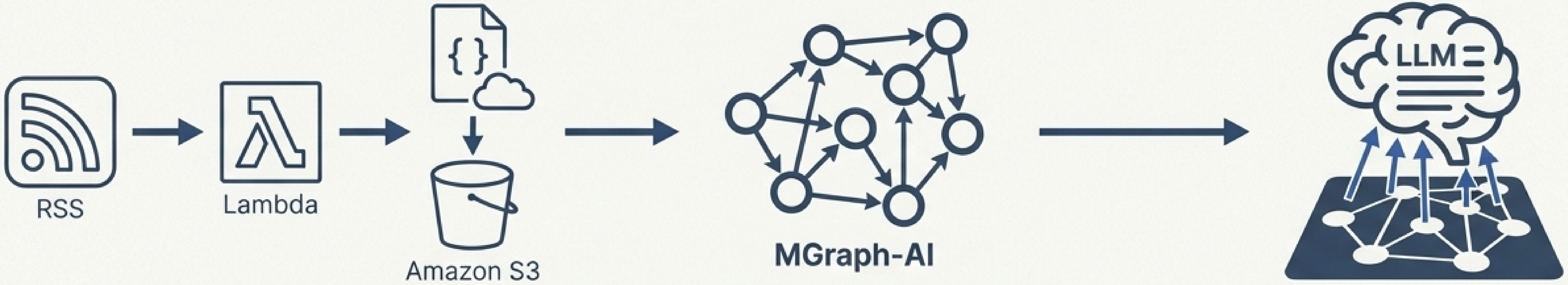
Project Overview: An MVP service that provides personalised news digests by transforming raw articles into a semantic knowledge graph before summarisation by an LLM.

How it Works: A Serverless Pipeline on AWS

Step 1: Ingest

Step 2: Structure

Step 3: Generate



AWS Lambda functions collect RSS feeds and store articles as structured JSON on S3.

MGraph-AI loads the JSON data into an in-memory graph, mapping entities and relationships between topics. This creates the semantic context.

The structured graph data is fed to an LLM, guiding it to produce summaries that are deterministic and traceable back to source facts.

> Demonstrates how an ephemeral graph database can ground LLMs in factual context, reducing 'hallucinations' and improving the quality and reliability of AI-generated content—a critical requirement for enterprise adoption.

A Portfolio Directly Aligned with AWS Strategic Priorities

Drive Serverless Leadership

OSBot-AWS · The Cyber Boardroom · MyFeeds.ai

Demonstrates complex, production-grade applications built with a serverless-first philosophy, providing powerful exemplars for other customers.

Accelerate GenAI & Amazon Bedrock Adoption

The Cyber Boardroom · MyFeeds.ai

Provides a compelling, real-world success story for Bedrock in the high-value cybersecurity domain. Showcases patterns for building trustworthy, explainable AI.

Fill the Graph Database Technology Gap

MGraph-AI

Offers a ready-made solution and architectural pattern for the missing lightweight, embedded graph engine for Lambda and Fargate.

Deepen Open Source & Developer Engagement

OSBot-AWS · MGraph-AI

Fosters community goodwill and provides mature, permissively licensed projects for AWS to sponsor, contribute to, and integrate with.

A Blueprint for a High-Impact Partnership

To capitalise on this strategic alignment, we propose a multi-faceted partnership designed to accelerate innovation, create powerful market narratives, and deliver new capabilities to AWS customers.

1. Co-Develop and Innovate

Actively partner on the evolution of MGraph-AI into a first-class offering for the AWS ecosystem.

2. Amplify and Showcase

Feature The Cyber Boardroom and other projects as official AWS case studies and thought leadership content.

3. Sponsor and Integrate

Provide resources to support open-source development and drive deeper integration with AWS services.

4. Package and Launch

Collaborate on go-to-market initiatives to bring these solutions to the AWS Marketplace and enterprise customers.

Co-Developing a Serverless Graph Offering and Creating Flagship Case Studies

Section 1: Co-Development

Partner with Dinis Cruz to sponsor and evolve MGraph-AI into an AWS-compatible managed library or service.

Co-author an AWS Solutions Implementation or Quick Start for MGraph-AI; integrate its concepts into a future first-party AWS service.

Success Metric

Within 12 months, a publicly announced AWS reference architecture or beta service for a "Serverless Graph DB on AWS" is released.

Section 2: Thought Leadership

Produce official AWS case studies, blog posts, and re:Invent talks on The Cyber Boardroom and MyFeeds.ai.

Highlight how serverless and Amazon Bedrock enable rapid, secure innovation in the AI era.

Success Metric

Publication of 2–3 official AWS case studies or blog posts within the next year, plus speaking slots at major AWS events.

Sponsoring Open-Source Integration and Launching Joint Solutions

Section 1: Open-Source Sponsorship

Provide funding, AWS credits, and technical resources to accelerate the development of OSBot-AWS and MGraph-AI.

Example Actions

Fund new features in OSBot-AWS to cover the latest AWS APIs; have AWS engineers contribute to MGraph-AI's integration with services like Neptune or SageMaker.

Success Metric

Measurable advancements in the open-source projects (e.g., new releases with AWS-optimised features) within 6-12 months.

Section 2: Joint Go-to-Market

Package solutions like The Cyber Boardroom architecture for the AWS Marketplace or as an AWS Quick Start solution.

Target Market

Enterprise clients in finance, healthcare, and other sectors where board-level cyber advisory is in high demand.

Success Metric

At least one joint solution or Marketplace offering launched, with pilot engagements secured with key AWS customers.

A Partnership for Mutual Strategic Gain

Summary of Benefits for AWS

- **Fast-Track Innovation:** Gain a head-start on a unique, in-demand serverless graph database offering.
- **Enrich the GenAI Portfolio:** Acquire authentic, enterprise-ready success stories for Amazon Bedrock.
- **Inspire Customers:** Use compelling case studies to demonstrate the art of the possible on AWS.
- **Differentiate in a Competitive Market:** Show that AWS not only provides tools but actively collaborates with pioneers to push boundaries.

By investing in this partnership, AWS can turn an individual innovator's dedicated, multi-year achievements into a broader strategic win. This collaboration will reinforce AWS's reputation as the premier cloud for cutting-edge development and create a new success story that resonates with technology and business leaders worldwide.